

Rethinking Power Safety: The inherent Benefits of 800V Low Voltage DC Architectures

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Agenda

- DC Arc Flash
- Converters Type
- Distributed LVDC architecture
- Why are unidirectional racks important?
    - Incident Energy Impact
- Fault Clearing Timings
- Present Industry Standards
- Call for Action

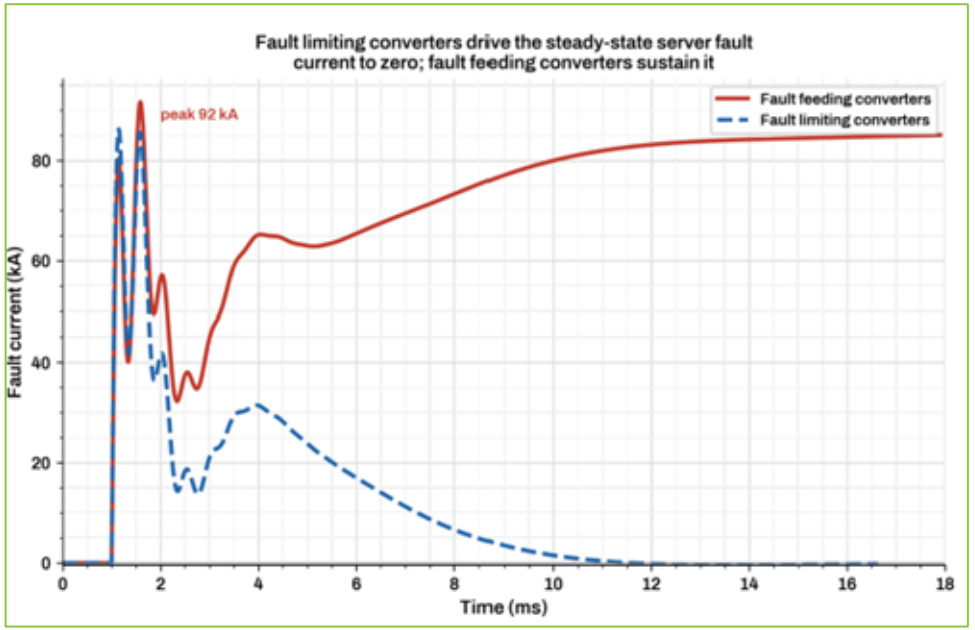
DC Arc Flash

- DC Arc Flash incident energy depends on
  - Prospective fault current
  - Fault clearing time

- The major influencing factors for the prospective fault current are:
  - **Power Converters' behaviour** to downstream fault.
  - **LVDC architecture**: has a great influence on the fault current.
  - **Unidirectional Racks**: Reverse current blocking diode at the rack level are a must.

- Major influencing factors for the Fault clearing time are:
  - **Protective device type**: Electromechanical breakers, fuses and Solid State Circuit Breakers (SSCBs) influence the response time.
  - **Fault location**: farther the fault location from protective devices, longer is the time to clear it .

Power Conversion Type

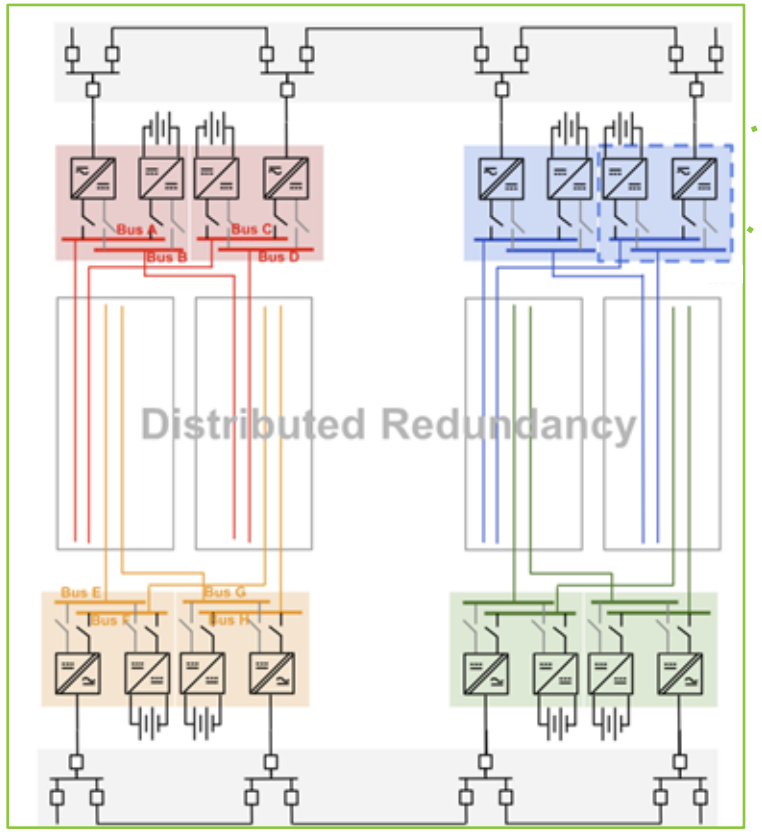


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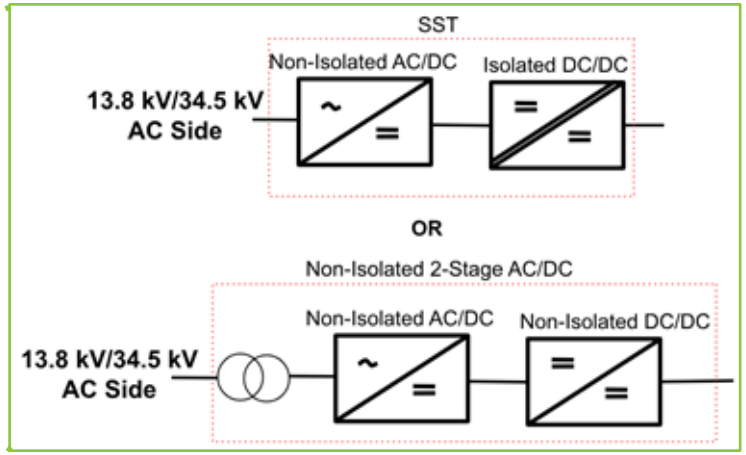
- Power Converter type plays a crucial role in the total fault current and arc flash energy
- **Fault limiting converters**, such as buck-type converters, can drive the fault current to zero - limiting the total fault energy.
- **Fault feeding converters,** like boost converters, sustains the fault current - leading to significantly higher fault energy.
- The initial peak of 80-90kA is contributed by the output capacitors of power converter, and is a direct function of the value of output capacitance.

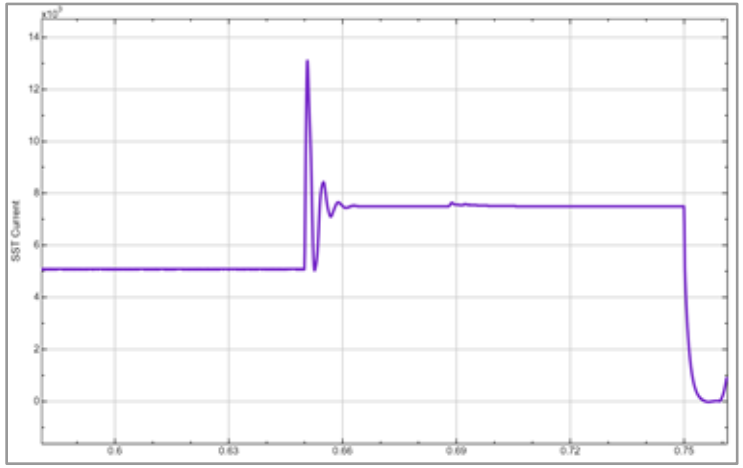
* Snapshot from DCF Power Distribution LVDC WhitePaper version 1.1:

Distributed LVDC Architecture



A Distributed redundancy architecture example with distributed energy storage in parallel with each source

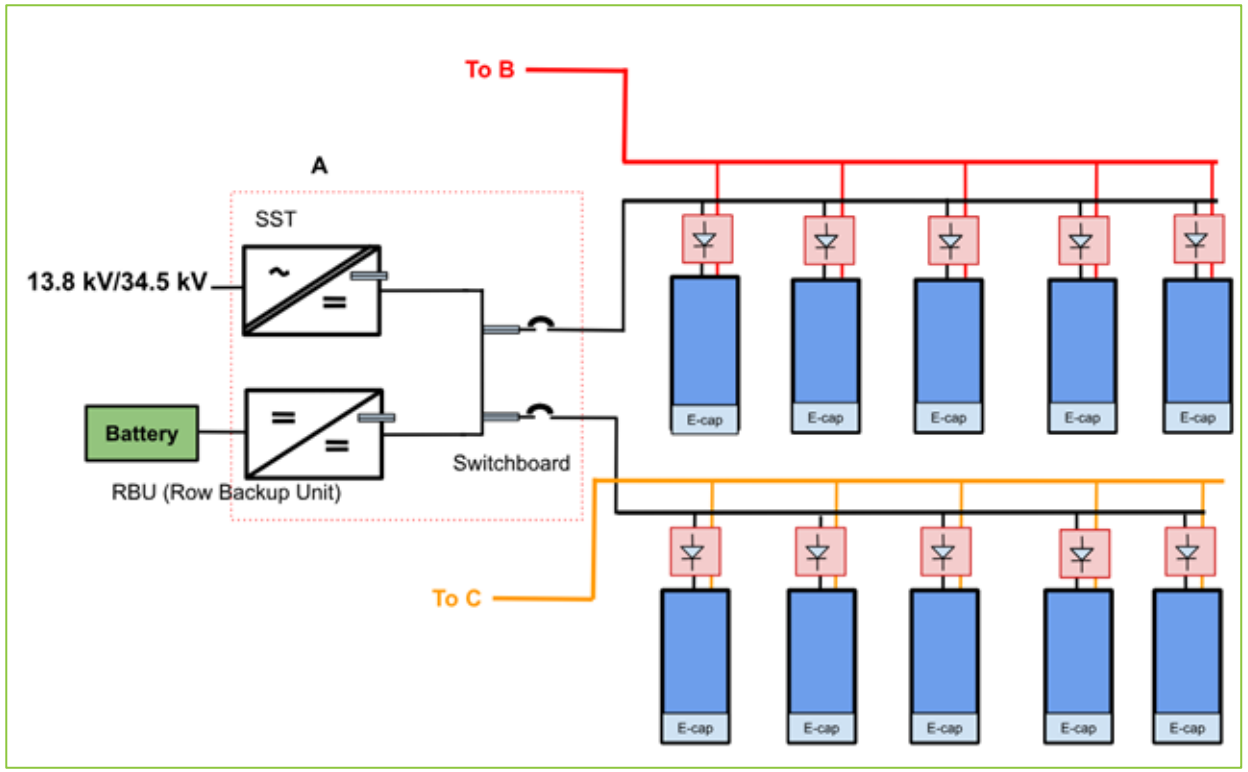




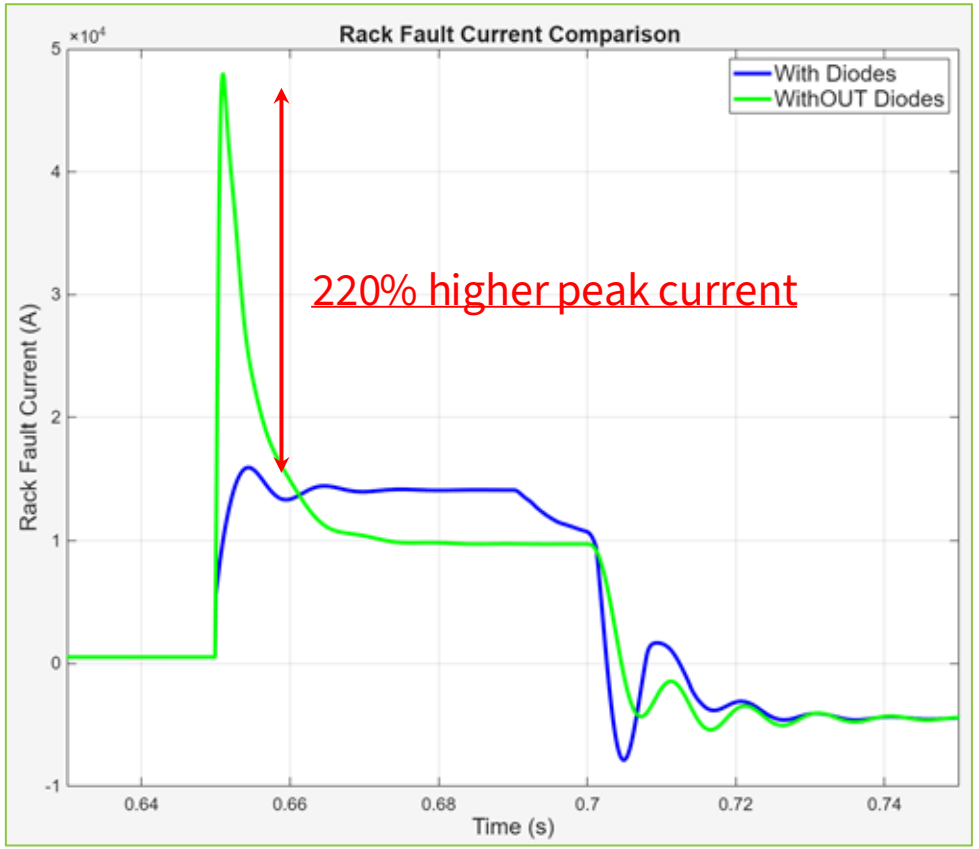
Example SST fault current waveform

Distributed energy sources coupled with fault current-limited power converters at source could reduce the prospective fault current.

Why are Unidirectional Racks Important?



Proposal Architecture with diodes either at rack level or Tap box level

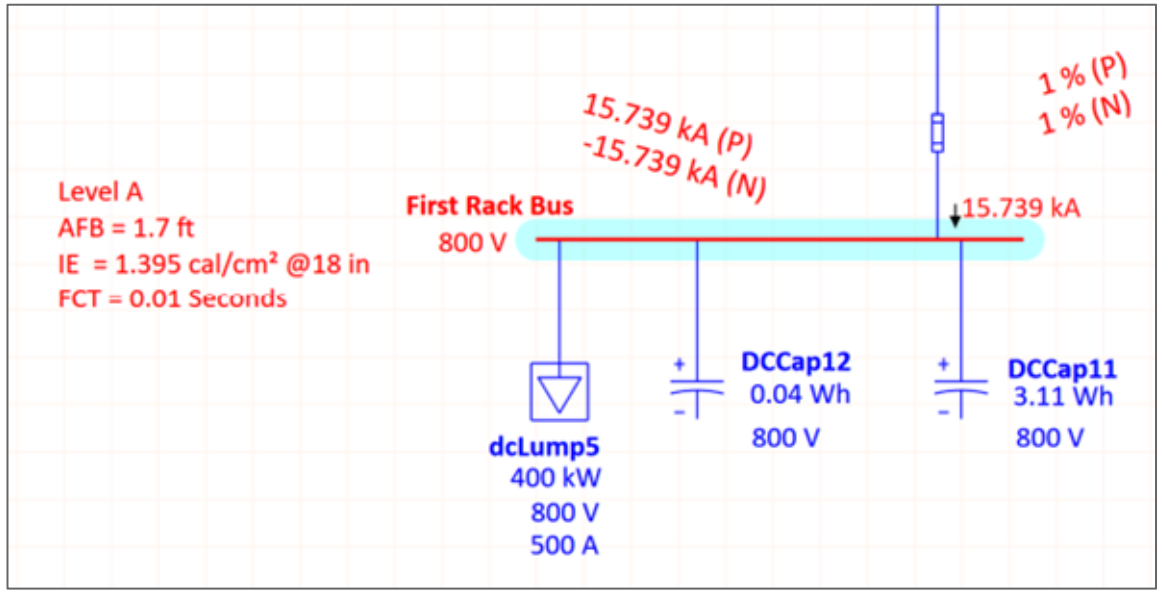


Simulated Rack fault currents highlighting the impacts of diodes in the fault current amplitude

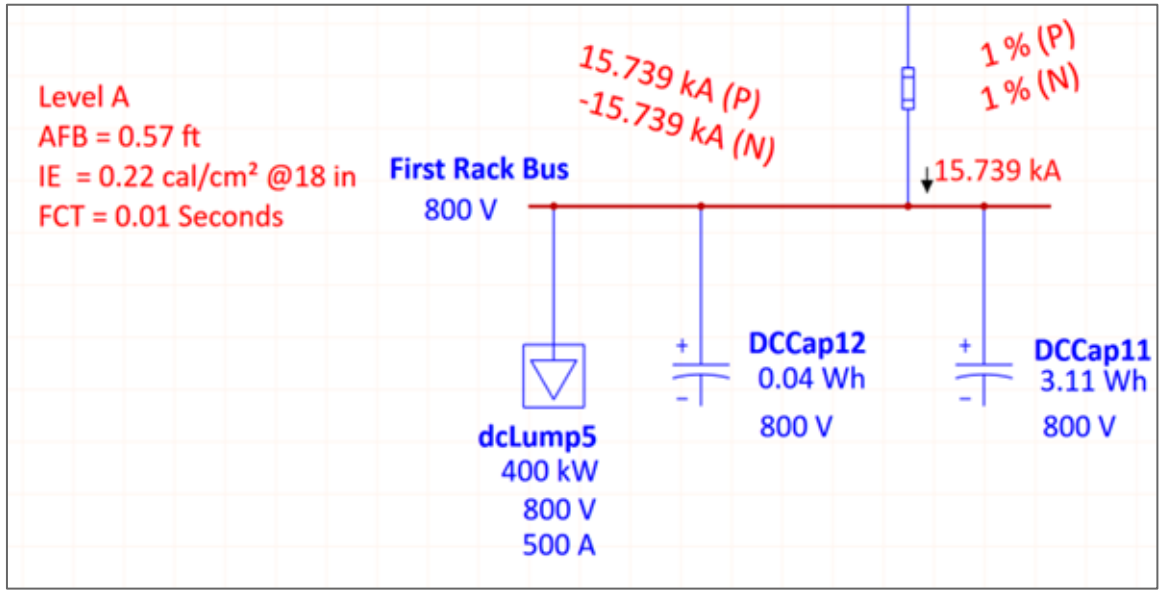
Diodes or MOSFETs (in synchronous rectification mode) necessary at rack or tap box level to limit the peak amplitude of fault current.

Why Unidirectional Racks: Incident Energy Impact

Without Diodes

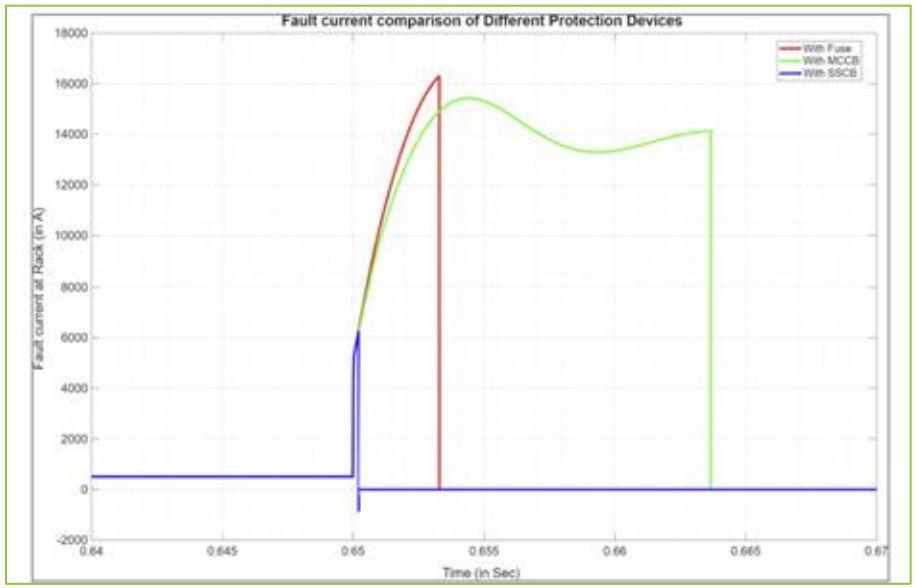


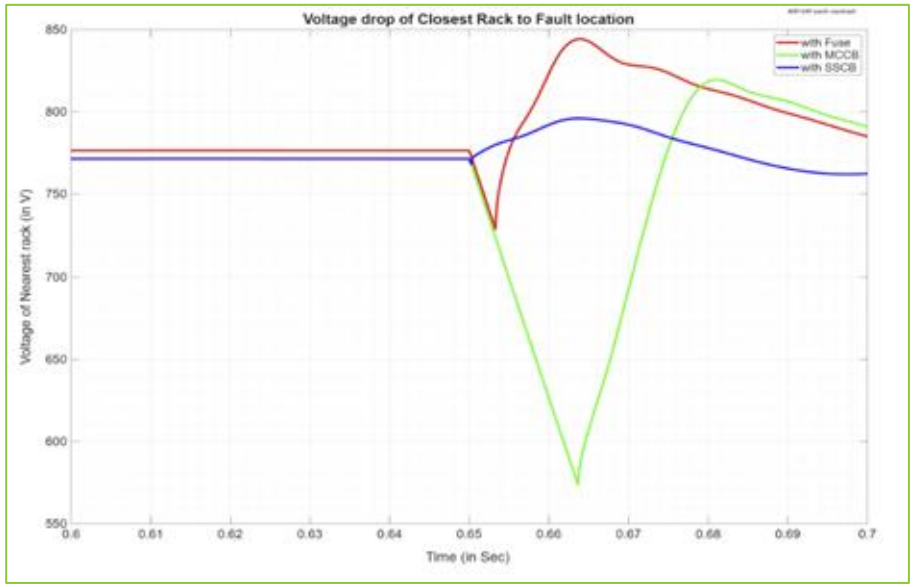
With Diodes



Reverse blocking diodes prevents the fault current backfeed, greatly reducing the peak current and the incident energy levels.

Fault Clearing Timings: Impact of Protective Devices





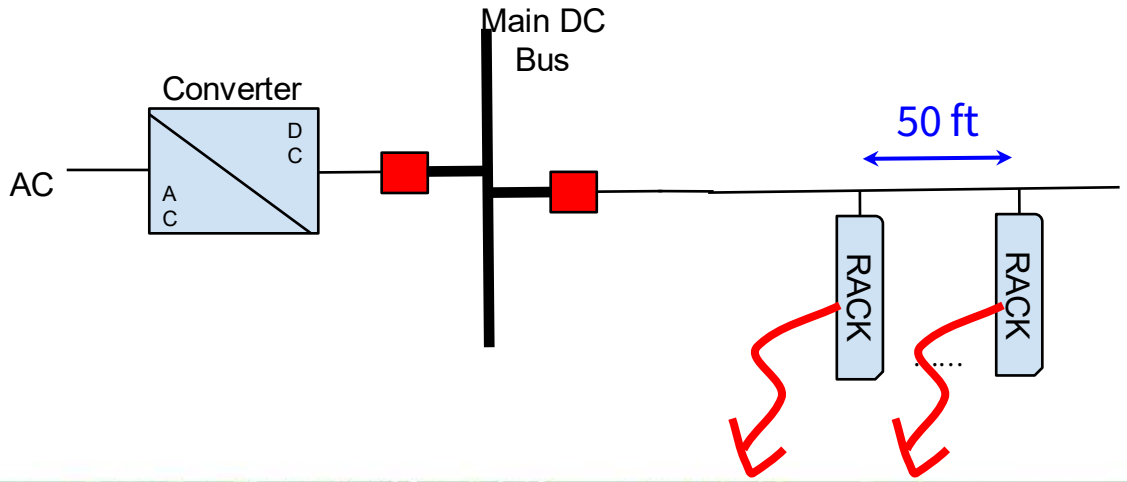
At the rack level	With Fuse	With Molded Case Circuit Breaker (MCCB)	With Solid State Circuit Breaker (SSCB)
	With diodes	With diodes*	With diodes
Fault current seen by protective device at the time of breaking (kA)	16 kA	15 kA	6.2 kA
Protective Device Response Time (sec)	3.3 ms	13 ms	0.2 ms
Cal/cm2	0.202 Cal/cm2	0.437 Cal/cm2	0.128 Cal/cm2

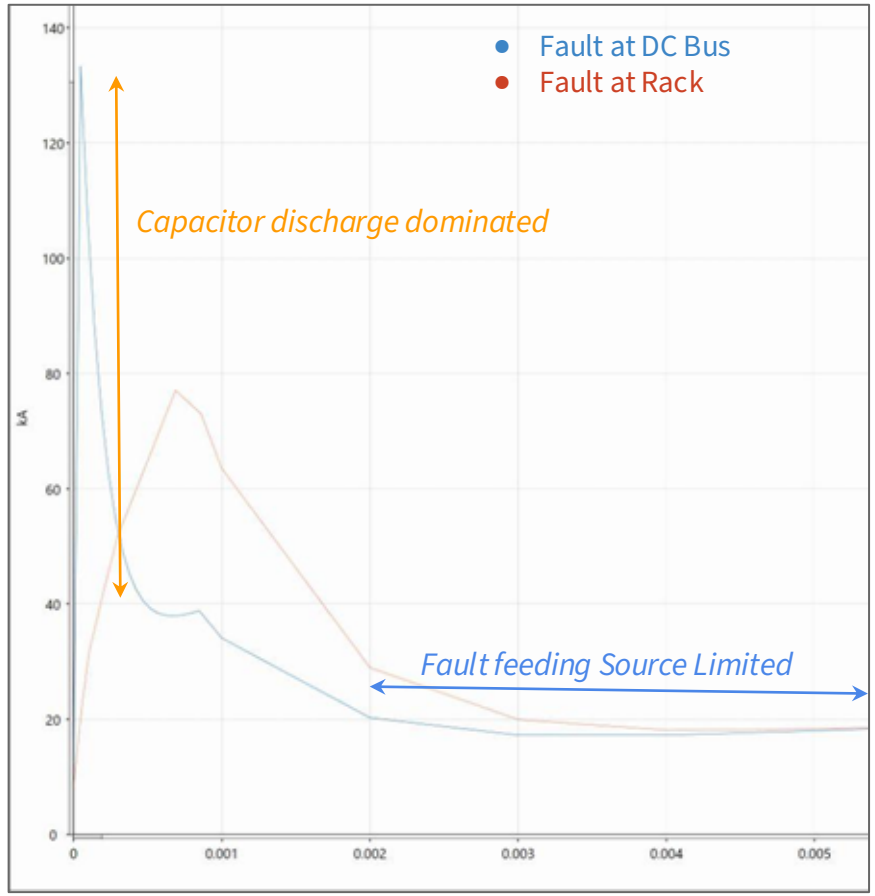
*Without diodes, the incident energy is >1.2Cal/cm2 with MCCB

Present Industry Standards

- IEEE 1584 does not cover DC systems.

- NFPA 70E addresses categories for energized DC system such as PhotoVoltaic (PV) arrays, battery banks but not converter-fed system.
  - Existing models overestimate arc flash energy in capacitor-dominated DC systems because they assume sustained maximum power, rather than decaying transient currents.





OCP Work

- DCF - Low Voltage Direct Current Power Distribution briefly covers the work on arc flash.
- Regular discussions are happening in the Power Distribution forum.

Call For Action

- Calling for joining collaborative efforts to drive the DC Arc Flash standards and methods (NFPA / IEEE) to take into effect converter-fed 800V dc architectures.
  - Combined effect of Capacitor Discharge and the converter limited current to be included in the calculations.
  - Extensive testing with various protective devices is required to understand how arc voltage affects fault clearing time and arc flash energy levels.

Thank You!

